MOHAMMED FERWANA

Backend Engineer · Node.js, Express, MongoDB

Email: mohammedferwana2@gmail.com • **Location:** Gaza, Palestine (Remote / Async Ready) • **GitHub:** github.com/hammoudFerwana • **LinkedIn:** linkedin.com/in/mohammed-ferwana

PROFESSIONAL SUMMARY

Methodical and reliability-driven Backend Engineer with demonstrated ownership in architecting high-integrity server-side systems, finite state machine (FSM) workflows, and reliable RESTful APIs. Experience leading backend engineering delivery, designing relational and document data schemas (PostgreSQL & MongoDB), enforcing strict role-based access control (RBAC), and authoring comprehensive automated integration test suites. Solid foundation in Computer Systems Engineering principles, algorithm design, and database systems.

TECHNICAL SKILLS & COMPETENCIES

Backend Runtimes:	Node.js, Express.js, RESTful API Design, Modular Architecture
Databases & Storage:	MongoDB Atlas, PostgreSQL, Mongoose, Compound Indexing, Data Modeling
Architecture & Security:	Finite State Machines (FSM), Multi-Tenant Isolation, JWT, RBAC, Data Validation, Rate Limiting
Testing & Quality:	Jest, Supertest, In-Memory DB Replicas, Integration Testing, Continuous Integration (CI)
DevOps & Tools:	Docker, Gitflow, GitHub Actions, Postman, Linux / Bash, Clean Architecture

PROFESSIONAL EXPERIENCE & TRAINING

Backend Developer Trainee — Aristo Software (Khireej Qader Program) — May 2026 – Present · Remote

- Delivering modular backend endpoints and business validation logic following clean architecture principles.
- Designing and optimizing MongoDB queries and data models for internal tooling.
- Participating in technical code reviews, documentation, and agile team workflows.

Backend Developer & Team Leader — TAQAT (TeamLine Platform) — 2026 · Remote

- **Architecture & Delivery:** Spearheaded backend system architecture for TeamLine, a collaborative project management system, ensuring predictable RESTful API contracts across teams.
- **Security & Access Control:** Designed secure authentication with JWT and granular workspace-scoped Role-Based Access Control (RBAC).
- **Engineering Leadership:** Conducted sprint planning, technical task estimation, and peer code reviews, accelerating engineering velocity and code consistency.
- **System Reliability:** Established centralized error handling pipelines, request sanitization, and database indexing that eliminated duplicate records.

Backend Engineering Fellow — Masar Tech — 2026 · Remote

- Accepted to Masar Tech Backend Engineering Bootcamp, Batch 2 (2026).
- Rigorous immersion in backend patterns, modular architecture, database design, and production testing.

Market Ready Developer Program — Gaza Sky Geeks (GSG) — 2023 – 2024 · Gaza, Palestine

- Completed rigorous professional immersion centered on full-stack architecture, clean code principles, and PostgreSQL relational database design.
- Architected reliable modular services adhering to separation of concerns, test-driven validation, and professional Gitflow branching workflows.
- Collaborated in pair programming, architectural reviews, and agile sprint cadence.

FEATURED ENGINEERING SYSTEMS & PROJECTS

Sawn (formerly InsurFlow) — Multi-Tenant B2B Motor Insurance Claim Platform — Sole Backend Engineer · 2026

- **Finite State Machine:** Designed and implemented a deterministic 10-state claim lifecycle state machine preventing invalid status mutations with explicit HTTP 409 conflict handling.
- **Multi-Tenant Compound Indexing:** Structured tenant-scoped data isolation in MongoDB Atlas using compound B-tree indexes with organizationId leading keys to guarantee zero data leakage.
- **Atomic Concurrency:** Engineered sequential claim numbering (`CLM-ORG-0001`) via atomic counter operations, verified collision-free under automated integration tests.
- **Automated Integration Testing:** Authored 616 automated integration tests across 39 test suites with Jest and MongoMemoryServer (run 2026-10-01), validating business invariants.

TeamLine — Collaborative Team Workspace Platform — Backend Developer & Team Leader · 2026

- Built modular REST API services for workspace membership, task assignment, and activity tracking.
- Enforced schema validation and indexing that sustained rapid query response times without duplication.

EDUCATION

B.Sc. in Computer Systems Engineering — Al-Azhar University (CGPA 93.3) — 2022 – Expected 2027

- Core Coursework: Operating Systems, Data Structures & Algorithms, Database Management Systems, Computer Networks, Software Engineering, Discrete Mathematics.